

I combined the methodology, results, and discussion of results.

Question 11.1

Using the crime data set `uscrime.txt` from Questions 8.2, 9.1, and 10.1, build a regression model using:

1. Stepwise regression
2. Lasso
3. Elastic net

For Parts 2 and 3, remember to scale the data first – otherwise, the regression coefficients will be on different scales and the constraint won't have the desired effect.

For Parts 2 and 3, use the `glmnet` function in R.

Notes on R:

- For the elastic net model, what we called λ in the videos, `glmnet` calls “alpha”; you can get a range of results by varying alpha from 1 (lasso) to 0 (ridge regression) [and, of course, other values of alpha in between].
- In a function call like `glmnet(x, y, family="mgaussian", alpha=1)` the predictors `x` need to be in R's matrix format, rather than data frame format. You can convert a data frame to a matrix using `as.matrix` – for example, `x <- as.matrix(data[,1:n-1])`
- Rather than specifying a value of `T`, `glmnet` returns models for a variety of values of `T`.

Stepwise Regression

- I. First, I performed backward elimination. Here is the output of my final model:

```
Step: AIC=503.93
Crime ~ M + Ed + Po1 + M.F + U1 + U2 + Ineq + Prob

      Df Sum of Sq    RSS   AIC
<none>                 1453068 503.93
- M.F   1     103159 1556227 505.16
- U1    1     127044 1580112 505.87
- Prob  1     247978 1701046 509.34
- U2    1     255443 1708511 509.55
- M     1     296790 1749858 510.67
- Ed    1     445788 1898855 514.51
- Ineq  1     738244 2191312 521.24
- Po1   1    1672038 3125105 537.93

Call:
lm(formula = Crime ~ M + Ed + Po1 + M.F + U1 + U2 + Ineq + Prob,
    data = uscrime)

Coefficients:
(Intercept)          M          Ed          Po1          M.F          U1
   -6426.10     93.32    180.12    102.65     22.34   -6086.63
          U2          Ineq          Prob
    187.35     61.33   -3796.03
```

Backward starts with the full model with all the variables and removes the variable one at a time to reduce the AIC. As a result, our final model consists of the variables: M, Ed, Po1, M.F, U1, U2, Ineq, and Prob. The AIC is 503.93, which is the lowest out of all of them.

II. Next, I performed forward elimination. Here is the output of my final model:

```
Step: AIC=504.79
Crime ~ Po1 + Ineq + Ed + M + Prob + U2
```

	Df	Sum of Sq	RSS	AIC
<none>			1611057	504.79
+ Wealth	1	59910	1551147	505.00
+ U1	1	54830	1556227	505.16
+ Pop	1	51320	1559737	505.26
+ M.F	1	30945	1580112	505.87
+ Po2	1	25017	1586040	506.05
+ So	1	17958	1593098	506.26
+ LF	1	13179	1597878	506.40
+ Time	1	7159	1603898	506.58
+ NW	1	359	1610698	506.78

```
Call:
lm(formula = Crime ~ Po1 + Ineq + Ed + M + Prob + U2, data = uscrime)
```

```
Coefficients:
(Intercept)      Po1      Ineq      Ed      M      Prob
-5040.50      115.02      67.65     196.47    105.02   -3801.84
      U2
      89.37
```

For forward selection, we start with the model with just the intercept and adds variables one at a time to reduce the AIC the most. The final model consists of the variables: Po1, Ineq, Ed, M, Prob, U2. The final model's AIC is 504.79.

- III. Finally, I performed stepwise regression (both directions). Here is the output of my final model:

```

Step:  AIC=503.93
Crime ~ M + Ed + Po1 + M.F + U1 + U2 + Ineq + Prob

      Df Sum of Sq  RSS   AIC
<none>                  1453068 503.93
+ Wealth  1      26493 1426575 505.07
- M.F     1     103159 1556227 505.16
+ Pop     1      16697 1436371 505.39
+ Po2     1      14148 1438919 505.47
+ So      1       9329 1443739 505.63
+ LF      1       4374 1448694 505.79
+ NW      1       3799 1449269 505.81
+ Time    1       2293 1450775 505.86
- U1      1     127044 1580112 505.87
- Prob    1     247978 1701046 509.34
- U2      1     255443 1708511 509.55
- M       1     296790 1749858 510.67
- Ed      1     445788 1898855 514.51
- Ineq    1     738244 2191312 521.24
- Po1     1    1672038 3125105 537.93

Call:
lm(formula = Crime ~ M + Ed + Po1 + M.F + U1 + U2 + Ineq + Prob,
    data = uscrime)

Coefficients:
(Intercept)          M          Ed          Po1          M.F          U1
   -6426.10      93.32     180.12     102.65      22.34    -6086.63
          U2          Ineq          Prob
     187.35      61.33    -3796.03

```

Stepwise elimination starts with the full model and looks at both additions and removals to the model at each step. The final model consists of the variables: M, Ed, Po1, M.F, U1, U2, Ineq, and Prob. The final model's AIC is 503.93.

As you can see, the AIC of the backward elimination and stepwise regression is the same (503.93). I concluded that stepwise regression is the better model because it considers both directions. However, it doesn't matter which one to choose since they have the same result.

Lasso

- I. First, I scaled the data.
- II. Next, I used `cv.glmnet()` to create my LASSO model. These are the parameters I chose to use:

```

```{r}
library(glmnet)
set.seed(123)
model_lasso = cv.glmnet(x = as.matrix(uscrime[, -16]),
 y = as.matrix(uscrime[, 16]),
 alpha = 1,
 nfolds = 8,
 nlambda = 20,
 type.measure = "mse",
 family = "gaussian",
 standardize = TRUE)
```

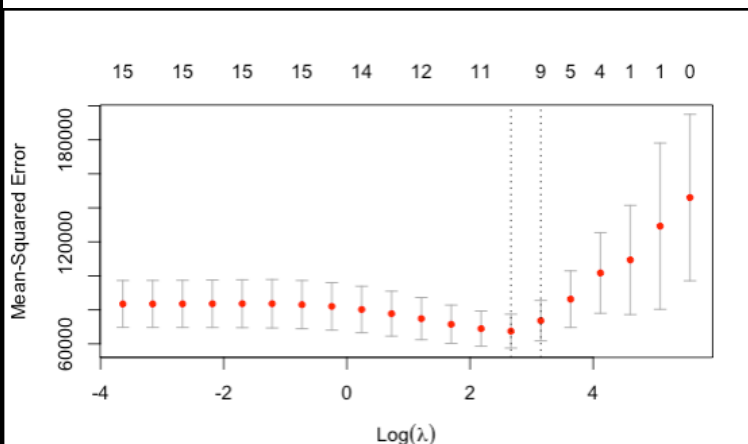
```

III. Next, I plotted the LASSO model and obtained the minimum lambda values and coefficients of the model.

```

s16  0.11263988  83568.64  14
s17  0.06936907  83510.97  15
s18  0.04272082  83476.84  15
s19  0.02630954  83471.15  15
16 x 1 sparse Matrix of class "dgCMatrix"
      s0
(Intercept) -4188.8210567
M             62.5173281
So            46.3065171
Ed            89.8715021
Po1          105.0118443
Po2           .
LF            80.0219314
M.F           15.7901041
Pop           .
NW            0.2279318
U1            .
U2           35.8812138
Wealth        .
Ineq          41.6921408
Prob        -3404.8060716
Time          .

```



As you can see in the plot above, the minimum MSE occurs at $\log(\lambda)$ when it is approximately equal to 2.2. At this λ , the model keeps 10 variables.

The right dashed line's λ is slightly larger. It also has a slightly larger MSE, and it uses 9 variables.

The output with the coefficients shows the non-zero variables (coefficients) of the λ .min. The coefficients are M, So, Ed, Po1, LF, M.F, NW, U2, Ineq, Prob. I will be focusing on prediction performance, so I will focus on using λ .min. These are the 10 variables.

IV. Finally, I created a function to calculate the MSE. Here is the output:

```
```{r}
min(model_lasso$cvm)
```

[1] 67492.91
```

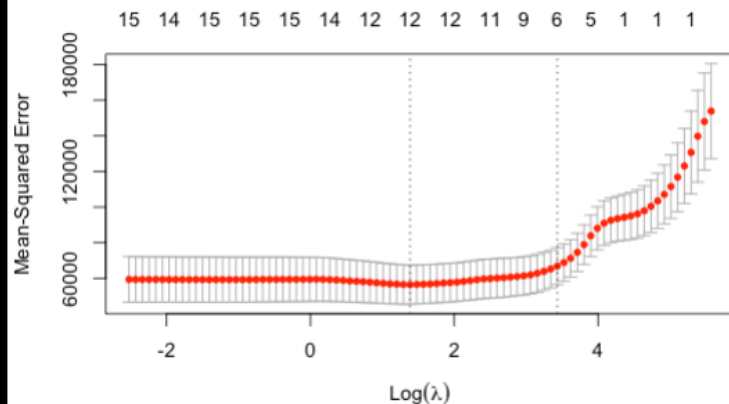
The MSE of the lambda minimum is 67492.91.

- V. I will try using a different lasso model. For this model, I will change nlambda to 100. The nlambda parameter controls the number of lambda values when fitting the model. In other words, we try 100 different values of lambda, ranging from a large value, which focuses strong regularization, to a small value, which focuses on weak regularization.

```
```{r}
model_lasso1 = cv.glmnet(x = as.matrix(uscrime[, -16]),
 y = as.matrix(uscrime[, 16]),
 alpha = 1,
 nfolds = 8,
 nlambda = 100,
 type.measure = "mse",
 family = "gaussian",
 standardize = TRUE)
```
```

- VI. Here are the plot and coefficients.

```
s84  0.10621207  59364.45  14
s85  0.09677648  59373.31  14
s86  0.08817912  59380.48  14
s87  0.08034553  59386.00  15
16 x 1 sparse Matrix of class "dgCMatrix"
      s0
(Intercept) -5.819456e+03
M            8.178332e+01
So           3.328593e+01
Ed           1.527383e+02
Po1          9.992079e+01
Po2          .
LF           .
M.F          1.798516e+01
Pop         -4.217949e-01
NW           1.295817e+00
U1          -3.727857e+03
U2           1.316352e+02
Wealth       4.968377e-02
Ineq         6.093690e+01
Prob        -3.889437e+03
Time         .
```



The lambda minimum uses 12 variables. The coefficients that are used are M, So, Ed, Po1, M.F, Pop, NW, U1, U2, Wealth, Ineq, and Prob.

VII. Here is the MSE.

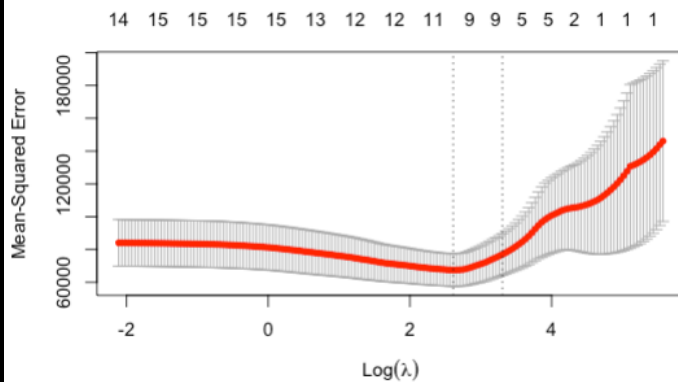
```
```{r}
min(model_lasso1$cvm)
```

[1] 56482.55
```

The MSE is 56482.55.

VIII. I created one last model where nlambda = 200. Let's see what changes.

```
s163  0.1392295  83991.01  14
s164  0.1329324  84009.05  14
s165  0.1269201  84026.05  14
s166  0.1211797  84043.30  14
16 x 1 sparse Matrix of class "dgCMatrix"
      s0
(Intercept) -4283.2724257
M             63.7512753
So            49.4389539
Ed            94.0054004
Po1           105.0455899
Po2           .
LF            93.9238193
M.F           15.8981244
Pop           .
NW            0.2285902
U1           -141.9173491
U2            40.6670522
Wealth       .
Ineq          42.3953795
Prob         -3456.9135044
Time         .
```



The lambda minimum uses 11 variables. The coefficients that are used are M, So, Ed, Po1, LF, M.F, NW, U1, U2, Ineq, and Prob.

IX. Here is the MSE:

```
```{r}
min(model_lasso2$cvm)
```

[1] 67491.38
```

The MSE is 67491.38.

Comparing all the models, the best model is the second model. The reason for this is that the MSE is the lowest for that model. Since I am valuing predictive performance, a lower MSE is ideal. Lower MSE means that the model has the smallest average prediction error. Additionally, the model uses only one more variable than the other two models.

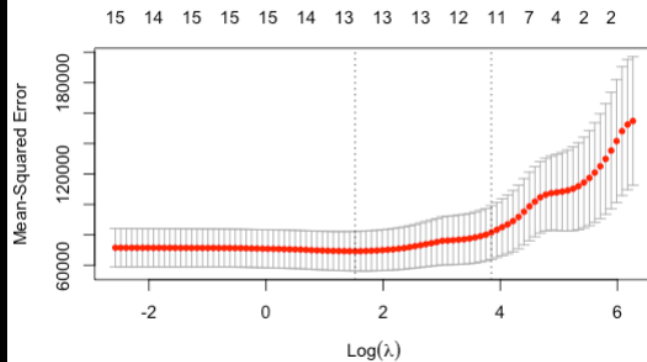
Elastic Net

- I. First, I created my Elastic Net model. To do this, I had to set alpha to 0.5. Here is the code I used:

```
```{r}
set.seed(123)
model_enet <- cv.glmnet(
 x = x_scaled,
 y = y,
 alpha = 0.5, # Elastic Net
 nlambda = 100, # Grid of 100 lambda values
 nfolds = 10, # 10-fold cross-validation
 type.measure = "mse", # Use MSE as performance metric
 family = "gaussian",
 standardize = TRUE # Already scaled manually
)
```
```

- II. Next, I found the coefficients and plot for the model.

```
s92 0.10091867 71475.98 15
s93 0.09195334 71486.52 15
s94 0.08378445 71497.33 15
s95 0.07634127 71506.53 15
16 x 1 sparse Matrix of class "dgCMatrix"
      s0
(Intercept) 905.08511
M           103.56327
So          16.90215
Ed          174.75228
Po1         288.70149
Po2         .
LF          .
M.F         55.49396
Pop         -19.20869
NW          19.02883
U1          -75.76496
U2          120.25967
Wealth      58.91639
Ineq        246.36956
Prob        -91.13275
Time        .
```



As you can see in the output, the minimum lambda uses 12 variables. The variables are M, So, Ed, Po1, M.F, Pop, NW, U1, U2, Wealth, Ineq, Prob.

III. I found the MSE of the model using `$cvm`.

```
{r}
min(model_enet$cvm)
[1] 69247.85
```

The MSE is 69247.85.

IV. I created a new elastic net model where I changed `nlambda` to 200 and `nfolds` to 15.

```

{r}
set.seed(123)
model_enet1 <- cv.glmnet(
  x = x_scaled,
  y = y,
  alpha = 0.5,          # Elastic Net
  nlambda = 200,        # Grid of 100 lambda values
  nfolds = 15,          # 10-fold cross-validation
  type.measure = "mse", # Use MSE as performance metric
  family = "gaussian",
  standardize = TRUE    # Already scaled manually
)

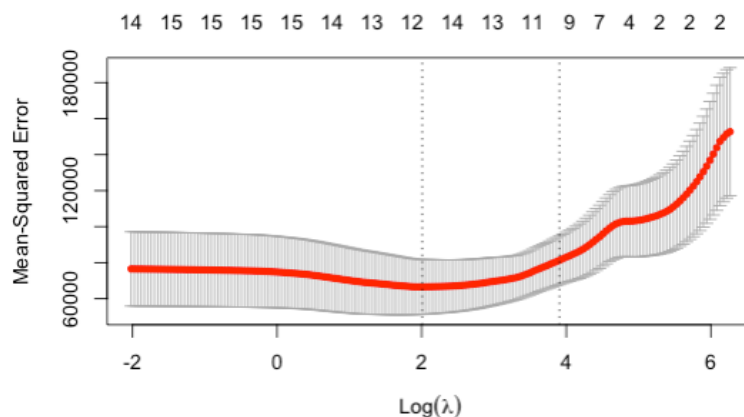
```

V. Here are the plot and coefficients.

```

s176 0.1525650 76465.02 14
s177 0.1456647 76487.36 14
s178 0.1390765 76508.00 14
s179 0.1327863 76524.65 14
16 x 1 sparse Matrix of class "dgCMatrix"
      s0
(Intercept) 905.08511
M            98.37010
So           19.63621
Ed           162.03243
Po1          279.52175
Po2           6.39211
LF            .
M.F           58.92884
Pop          -11.02706
NW            18.57127
U1           -67.17689
U2           107.90643
Wealth        46.73384
Ineq          225.30551
Prob         -89.66047
Time           .

```



The new model's lambda minimum uses 13 variables: M, So, Ed, Po1, Po2, M.F, Pop, NW, U1, U2, Wealth, Ineq, Prob.

VI. Finally, I calculated the MSE.

```
```{r}
min(model_enet1$cvm)
```

[1] 66538.81
```

The MSE is 66538.81.

The MSE of the new model is lower than the MSE of the first model. However, it uses one extra variable. In this case, we want to consider whether overfitting is an issue here. I think it is fine to include one extra variable and settle with the new model. Therefore, I think the new model is a better model to use since it has a couple thousands difference in the MSE.

References:

AI

<https://www.rdocumentation.org/packages/stats/versions/3.6.2/topics/step>

<https://www.rdocumentation.org/packages/glmnet/versions/4.1-10/topics/cv.glmnet>

https://mediaspace.gatech.edu/media/ISYE+6501+OMSA+Office+Hour/1_ui0s5fe1